# Rick Friedman

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## WORK EXPERIENCE

**CTRL+R,** Blacksburg, VA
**Software Engineer - Robotics [Contract]** Jan 2026 – Present

- Owned the end to end implementation, from technical designs to closing partnerships, of the browser-based robotics teleoperation platform for remote control, navigation, and fleet management of physical robots over WebRTC
- Matured early stage MVP into production grade offering by identifying and patching security vulnerabilities, improving reliability and safety, establishing robust data logging systems, and implementing automated resiliency
- Developed core features of the on-robot Rust agent (remote ROS2 commands, waypoint navigation, teleoperation safety) and a cloud-hosted Isaac Sim demo on AWS, enabling demos and customer evaluations
- Scaled platform compatibility across various hardware configurations including Unitree, Slamtec, Bracket Bot, and Booster

**Rovisys,** Raleigh, NC
**Systems Engineer** Aug 2025 – Jan 2026

- Developed a centerline system in Aveva System Platform for a bottling facility, standardizing production variables
- Co-authored the Functional Design Specification with the client, scoping requirements and control logic

**Virginia Tech Bio-Inspired Science and Technology Lab,** Blacksburg, VA
**Lead Undergraduate Researcher** Jan 2024 – May 2025

- Led the design and manufacturing of a soft robotics tendon actuation rig that mimics bat ear movements used for navigation
- Standardized the manufacturing process of robotic parts utilizing SLA and FDM 3D printing

**Lhoist North America,** Ripplemead, VA
**Mechanical Engineering Co-op** May 2023 – Dec 2023

- Led three major equipment modification projects that saved the company upwards of $3 million in building redesigns
- Orchestrated the inventorying, installation, and commissioning of a $100,000 emissions monitoring system
- Designed Python programs to analyze fluid flow, minimize piping network costs and improve environmental data analysis

**Robotic Perception,** Tel Aviv, Israel
**Robotics Engineering Intern** May 2022 – Aug 2022

- Modified and tested tractor vehicle mechanics to decrease manufacturing costs and better satisfy clients' criteria
- Produced electrical wiring diagrams and refined vehicle mechanical assemblies to support prototype builds and field testing

## PROJECTS

**Online Shop Owner,** Fall 2024-Present
**Rilos3DSolutions**

- Online shop used to sell 3D printed parts that I have designed, two years in operation with over $10,000 in revenue

**Virginia Tech Capstone Project,** Fall 2024-Spring 2025
**Development of an Autonomous Ground Rover**

- Implemented Nav2 path planning with Lidar/GPS/IMU/Depth sensor fusion in ROS2 for navigation in unmapped outdoor environments
- Designed the embedded electronics, battery, and motor systems for long-range power efficiency and reliability
- Developed a Gazebo simulation environment to validate path planning and obstacle avoidance before physical testing

## EDUCATION

**VIRGINIA TECH** Blacksburg, VA
**B.S. Mechanical Engineering** Graduated May 2025
**Concentration: Robotics and Mechatronics**

## SKILLS

**Proficient**: Rapid Prototyping, ROS2, Isaac Sim, Python, C++, Linux, SolidWorks, Fusion 360, MATLAB, 3D Printing, Rust, Nav2, SLAM, Docker, Git, CI/CD, AWS, Aveva System Platform